

BUVIJAG AI

Sharp Intelligence for a Data-Driven World.

Hackathon Project Documentation

BUVIJAG AI is an AI-powered business data analysis agent designed to turn uploaded business data into insights, visualizations, recommendations and an executive report.

1. Problem Statement

Businesses frequently have sales or operational data but need technical expertise to extract useful insights. BUVIJAG AI provides a simple upload-to-insight workflow.

2. Solution Workflow

Upload data → understand structure → assess data quality → detect KPIs and anomalies → generate visualizations → produce AI insights → recommend actions → simulate scenarios → export an executive report.

3. Core Features

Area	Capabilities
Data Quality	Data Health Score, missing values, duplicates, suspicious values
Business Analytics	Revenue, cost, profit, orders, AOV, quantity, trends, product/category/state/payment/review/delivery anal
AI Insights	Evidence-backed insights, executive summary, anomaly explanations
Decision Support	What Should I Do Next?, What-If Business Simulator
Inventory	Critical/reorder, low stock, stock-OK, supplier and inventory quality checks
Reporting	Executive PDF report
UX	Animated welcome page, four themes, Agent Activity, Recent history

4. Technology Stack

Python, Streamlit, Google Gemini API (google-genai), Pandas, NumPy, Matplotlib, Plotly, ReportLab, pypdf, Pillow, OpenPyXL and xlrd.

5. Security

API keys are stored outside the repository using Streamlit secrets. No secret key should be committed to GitHub.

6. Demo Data

The included demo dataset is synthetic and intentionally contains controlled data-quality and inventory scenarios to demonstrate the product. It must not be presented as real customer data.

7. Hackathon Demo Sequence

Upload CSV → Data Health → KPIs → Recommended Visualizations → AI Insights → Anomalies → Inventory Intelligence → What-If Simulator → Action Plan → Executive PDF → Recent.

8. Product Positioning

BUVIJAG AI acts as an AI business intelligence assistant that helps a user move from raw data to understandable evidence and practical next steps.